Omar Abdul Qadir

Embedded Software Engineer

Contact Info:

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Summary:

Senior Embedded Software Engineer who thinks about hardware from a software perspective with 3+ years across firmware, RTOS-based systems, and Embedded Linux platforms for smart-grid and communication applications. Proven track record spanning low-level driver development and hardware bring-up to leading device architecture, Linux system builds, and manufacturing test processes. Focused on building reliable, high-performance systems under real-world constraints, with a growing interest in Embedded ML for resource-constrained devices.

Work Experience:

Senior Embedded Software Engineer

Pylon Nov 2023 – present

- Stepped into the Senior role, Supporting device manufacturing and leading development of board test jigs on the factory floor.
- Architected a new Linux-based DCU (PLC base-node) from the ground up; application structure, build system, web/PLC network manager, and device DB, managing edge data for 1000+ smart grid nodes.
- Built the Linux system from the ground up; U-Boot bootloader, kernel configuration, and root filesystem, cutting boot time from 4 minutes (other competitors) to 45 seconds and lowering the system's memory footprint to 3GB.
- Led bring-up of a new advanced Gateway device (Ethernet Copper & Fiber) across multiple custom PCB iterations, cutting development cycles by 60% through close hardware-software coordination.
- Transitioned Gateways from bare-metal to RTOS (FreeRTOS), boosting CPU idle time by ~90%.
- Built a secure bootloader to safeguard devices against corrupted or unauthorized updates.
- Designed low-level EEPROM/Flash memory management, preventing data corruption under heavy load.
- Brought up the PLC (PRIME 1.3.6) and DLMS-HDLC stacks, reaching 99% data transmission reliability across smart meter networks.

Embedded Systems Part time Instructor

Smart Technology Jun 2025 – Present

- Authored 20+ practical challenges for the ATmega32 MCU, covering GPIO, Timers, PWM, and External Interrupts.
- Delivered instruction and C/C++ code reviews to nearly 60+ students, strengthening their bare-metal architecture understanding.

Professional Development:

Machine learning for Embedded Systems
Udemy June 2026

- Deployed end-to-end TinyML applications on resource-constrained microcontrollers, covering the full pipeline from model training to on-device inference.
- Applied hardware-aware model optimization techniques for Edge AI, targeting Cortex-M class processors and dedicated neural processing units.
- Applied model compression techniques, including quantization, weight pruning, and clustering, to run memory-bound deep learning models within tight embedded hardware constraints.

The Complete Prompt Engineering for AI Bootcamp (2026)
Udemy April 2026

- Applied prompt design techniques, including zero-shot, few-shot, and chain-of-thought prompting, to improve LLM performance and accuracy.
- Developed strategies for applying generative AI tools to real-world problem-solving through iterative prompt tuning and role-based prompting.
- Applied AI ethics and risk-mitigation practices to reduce model bias and hallucinations.

Embedded Linux
EDGES September 2024

- Built Embedded Linux systems using Yocto, Buildroot, and manual build processes.
- Developed and deployed Linux applications in C++ using OOP principles, with custom Makefile and CMake build systems.
- Wrote Bash scripts for automation, system setup, and debugging.
- Built custom embedded Linux environments, integrating drivers, services, and user-space components..

Education:

Bachelor Degree in Systems and Biomedical Engineering
El Shorouk Academy, Cairo, Egypt 2015 – 2020

Core Skills:

Programming Languages: C / Embedded C (primary), C++ (OOP, Linux applications), Python, Bash.
Communication Protocols: SPI, I2C, UART, CAN, LIN, PLC (Power Line Communication), Ethernet, Wireless (BLE, WI-FI, Zigbee).
Embedded Systems: RTOS, Embedded Linux, Bare-metal, Device Drivers, Bootloaders, ARM Cortex-M, Firmware Development, Memory Management, Hardware Bring-up.
Tools: Yocto, Buildroot, GDB, JTAG, SWD, Logic Analyzer, Oscilloscope, Git, Makefile, CMake, CI/CD.

Language Skills:

- Arabic: Native speaker.
- English: Advanced level.